

Midterm

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Problem 1. Let G be a group. Exhibit (with proof) G as the group given by some set of generators and relations.

Proof. Let G be a group. Let $S = G$. Thus $\forall g \in G, s_1 = g \in S, k = 1$ and $g = s_1^{\pm 1} = g$. Thus there exists a subset $S \subseteq G$, such that $\forall g \in G, g = s_1^{\pm 1} \dots s_k^{\pm 1}$ for $k \in \mathbb{Z}$ and $s_i \in S$ for $i \in 1 \dots k$. Let $X \subseteq G$ be the smallest such subset satisfying the condition above. Thus $\forall g \in G, g = s_1^{\pm 1} \dots s_k^{\pm 1}$ for $k \in \mathbb{Z}$ and $s_i \in X$ for $i \in 1 \dots k$. $X = \{x_1, \dots, x_n\}$. Let $s : X \rightarrow F(X)$ be a function which takes $x_i \rightarrow x_i$ for all $i \in 1 \dots n$. Let $t : X \rightarrow G$ be a function which takes $x_i \rightarrow x_i$. By the universal property of free groups, there exists a unique homomorphism $f : F(X) \rightarrow G$ such that $t = f \circ s$. Thus $f(x_i) = x_i$ for all $i \in 1 \dots n$. Let $K = \ker f$. By the definition of the subset $X, \forall g \in G, g = s_1^{\pm 1} \dots s_k^{\pm 1} = f(s_1^{\pm 1}) \dots f(s_k^{\pm 1})$ for $k \in \mathbb{Z}$ and $s_i \in X$ for $i \in 1 \dots k$. $X = \{x_1, \dots, x_n\}$. Since f is a homomorphism, $g = f(s_1^{\pm 1}) \dots f(s_k^{\pm 1}) = f(s_1^{\pm 1} \dots s_k^{\pm 1})$. Let $s = s_1^{\pm 1} \dots s_k^{\pm 1}$ and $s \in F(X)$. Thus for all $g \in G$, there exists a $s \in F(X)$ s.t $f(s) = g$. Thus f is surjective. By the First Isomorphism Theorem, $F(X)/K \cong f_*(F(X))$. Since f is surjective, $f_*(F(X)) = G$ and thus $F(X)/K \cong G$. By the definition of generators, X is a set of generators. Furthermore, by the definition of relations, K is the set of relations on G . Since $F(X)/K \cong G$, G is given by a set of generators X and a set of relations K . $\square$

Problem 2. Let X and Y be subgroups of G , with X contained in Y . Show that the natural map $G \rightarrow G/Y$ factors through the natural map $G \rightarrow G/X$, and that every fibre of the induced $G/X \rightarrow G/Y$ is bijective to Y/X . Conclude that if Y/X and G/Y are both finite, then G/X is finite and $|G/X| = |G/Y||Y/X|$ (do not assume any of X, Y or G is finite).

Proof. Let G be a group. Let X and Y be subgroups of G s.t $X \subseteq Y \subseteq G$. Let $p : G \rightarrow G/X$ be the natural quotient map. Let $q : G \rightarrow G/Y$ be the natural quotient map. $\forall a, b \in G$ s.t $p(a) = p(b)$, by the definition of $p, aX = bX$. Thus $b \in aX$ and by the definition of a left coset there exists a $x \in X$ s.t $b = ax$. Since $X \subseteq Y, x \in Y$, thus $ax \in aY$ and $b \in aY$. Thus $aY = bY$ and $q(a) = q(b)$. Thus $\forall a, b \in G, p(a) = p(b) \implies q(a) = q(b)$. Thus by the prehomework theorem there exists a $f : G/X \rightarrow G/Y$ s.t $q = f \circ p$. Moreover since p, q are surjective and $q = f \circ p$, f must be a surjective by properties of composition of function and must be unique by the prehomework.

$\forall g \in G$, let $\phi : Y/X \rightarrow f^*({gY})$ which is defined as the following $\forall yX \in Y/X, \phi(yX) = gyX$. $\forall yX, aX \in Y/X$ s.t $yX = aX$ where $y, a \in G$. Thus $a \in yX$ and there exists a $x \in X$ s.t $a = yx$. $\phi(yX) = gyX$ and $\phi(aX) = gaX$. Since $a = yx, gaX = gyxX$, since X is a group, $x^{-1} \in X$ thus $gyxx^{-1} = gy \in gyxX = gaX$. Thus $gy \in gaX$, and $\phi(yX) = gyX = gaX = \phi(aX)$. Thus for all $yX, aX \in Y/X$ s.t $yX = aX$ then $\phi(yX) = \phi(aX)$. Thus ϕ is well defined.

1. **Injective:** $\forall yX, zX \in Y/X$ s.t. $\phi(yX) = \phi(zX)$. By the definition of ϕ , $gyX = gzX$, thus $gz \in gyX$ and, by the definition of a coset, $\exists x \in X$ s.t. $gz = gyx$ thus $g^{-1}gz = g^{-1}gyx$ or $z = yx$. Since $x \in X$, by the definition of a left coset, $z \in yX$ and $yX = zX$. Thus ϕ is injective.
2. **Surjective:** $\forall aX \in f^*(\{gY\})$, by the definition of a preimage, $f(aX) = gY$. Thus $f(p(a)) = q(a) = aY = gY$. Thus $a \in gY$. Thus there exists a $y \in Y$ s.t. $a = gy$. Thus $aX = gyX$ and $yX \in Y/X$ and $\phi(yX) = gyX = aX$. Thus $\forall aX \in f^*(\{gY\})$, $\exists yX \in Y/X$ s.t. $\phi(yX) = aX$. Thus ϕ is surjective.

Since ϕ is surjective and injective, it is bijective. Thus there exists a bijection from Y/X to $f^*(\{gY\})$ for all $g \in G$, thus there exists a bijection from Y/X to every fibre of f .

Suppose Y/X and G/Y are both finite. Since there is a bijection from every fibre of f to Y/X , each fibre of f must be of the same order thus $\forall g \in G$, $|f^*(\{gY\})| = |Y/X|$. $|G/Y|$ is the number of fibres of f . $\forall g, h \in G$, s.t. $aX \in f^*(\{gY\}) \cap f^*(\{hY\})$, thus by definition of preimage, $f(aX) = gY$ and $f(aX) = hY$. Thus $gY = hY$. Thus each fibre is pairwise disjoint or equal. Since f is a function, every element is in 1 fibre. Since every element is in 1 fibre, each fibre is pairwise disjoint and contains $|Y/X|$ elements, and there are $|G/Y|$ fibres, then $|G/X| = |Y/X||G/Y|$. Since Y/X and G/Y are both finite and the product of two finite values must be finite as well, $|G/X|$ is finite. $\square$

Problem 3a. Let N, H be subgroups of G , with H of finite index. Assume N is normal. Construct a natural bijection between $N/N \cap H$ and NH/H .

Proof. Let N, H be subgroups of G , with H of finite index. Assume N is normal. Let $f : N/N \cap H \rightarrow NH/H$ defined as follows, $\forall n(N \cap H) \in N/N \cap H$, $f(n(N \cap H)) := nH$.

For all $a(N \cap H), b(N \cap H) \in N/(N \cap H)$ s.t. $a(N \cap H) = b(N \cap H)$ where $a, b \in N$. $b \in a(N \cap H)$ and $\exists h \in N \cap H$ s.t. $b = ah$. $f(a(N \cap H)) = aH$ and $f(b(N \cap H)) = bH$. Since H is a subgroup, $e \in H$, and $b = be \in bH$ by definition of a left coset. Since $h \in N \cap H$, $h \in H$, thus $b = ah \in aH$ by definition of left coset. Since $b \in bH \cap aH$, $bH = aH$, since left cosets are pairwise disjoint or equal. Thus $f(a(N \cap H)) = f(b(N \cap H))$, meaning f is well defined.

1. **Injective:** For all $a(N \cap H), b(N \cap H) \in N$ s.t. $f(a(N \cap H)) = f(b(N \cap H))$ where $a, b \in N$. By the definition of f , $aH = bH$. Thus $b \in aH$ and by definition of a left coset, $\exists h \in H$ s.t. $b = ah$. Since N is a subgroup of G , $a^{-1} \in N$. Thus $a^{-1}b = h$. Since N is a subgroup of G , it is closed so $h \in N$, thus $h \in N \cap H$. Thus $b = ah \in a(N \cap H)$ and $a(N \cap H) = b(N \cap H)$. Thus f is injective.
2. **Surjective:** $\forall aH \in NH/H$, since $a \in NH$ it will be of the form $a = nh$ for $n \in N$ and $h \in H$. Thus $aH = nhH$. Since H is a subgroup of G , by definition of a subgroup $h^{-1} \in H$. Thus $nhh^{-1} \in nhH = aH$ thus $n \in aH$. Thus $aH = nH$. By the definition of f , $f(n(N \cap H)) = nH = aH$. Thus $\forall aH \in NH/H$, there exists a $n(N \cap H) \in N/N \cap H$ s.t. $f(n(N \cap H)) = aH$. Thus f is surjective.

Since f is surjective and injective it is bijective. Thus there exists a bijection between $N/N \cap H$ and NH/H . $\square$

Problem 3b. Construct a natural map from G/H to G/NH such that every fibre is bijective to $N/N \cap H$. Now assume H has finite index in G . Show that $N \cap H$ has finite index in N , $|N/N \cap H||G/NH| = |G/H|$. In particular the index of $N \cap H$ in N divides the index of H in G .

Proof. Let $q : G \rightarrow G/NH$ be the natural quotient map. Let $p : G \rightarrow G/H$ be the natural quotient map.

Since $H \subseteq NH \subseteq G$, by problem 2 there exists a $g : G/H \rightarrow G/NH$ s.t $q = g \circ p$. Since q and p are surjective, g must be surjective as well. Moreover by problem 2, each fibre of g is in bijection with NH/H . By problem 3a, there exists a natural bijection between $N/N \cap H$ and NH/H . Since there exists a bijection between each fibre of g and NH/H and a bijection between NH/H and $N/N \cap H$, there exists a bijection between each fibre of g and $N/N \cap H$.

Assume that H has finite index in G . Since each fibre of g is a subset of G/H and since G/H is finite each fibre of g must also be finite. Furthermore since each fibre of G is in bijection with $N/N \cap H$, $|N/N \cap H|$ must also be finite. $|G/NH|$ is the number of fibres of g . Furthermore more since each fibre of g are disjoint pairwise or equal and their union is G/H . Thus $|G/H| = |G/NH||N/N \cap H|$ since $|G/NH|$ is the number of fibres and $|N/N \cap H|$ is the size of each fibre. Since $|G/NH|$ is a positive integer, $|N/N \cap H|$ divides $|G/H|$. $\square$

Problem 4. Give an example of subgroups H and N of a finite group G such that the index of $H \cap N$ in N DOES NOT divide the index of H in G (so 3b does not hold without the assumption that N is normal).

Proof. Suppose $G = D_3 = \langle x, y \mid x^3 = y^2 = e, xy = yx^{-1} \rangle = \{e, x, x^2, y, xy, x^2y\}$ and $H = \{e, xy\}$ and $N = \{e, y\}$. $xyy(xy)^{-1} = xy^{-1}x^{-1} = xyx^{-1} = x^2y \notin N$. Thus N is not normal. $N \cap H = \{e\}$. $N/N \cap H = \{e(N \cap H), y(N \cap H)\}$ thus $|N/N \cap H| = 2$. $G/H = \{eH, xH, x^2H\}$ thus $|G/H| = 3$. $|N/N \cap H|$ does not divide $|G/H|$. Thus 3b doesn't hold without the assumption that N is normal. $\square$